Arunabha Majumder

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Portfolio: arunabha22.github.io/arunabha-majumder

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Yuman, DTU Skylab, Lyngby

Unsolicited application: Mechanical design / Robotics engineering

Dear Yuman team,

I am sending an unsolicited application for a role in mechanical design or robotics engineering. I am a mechatronics and robotics engineer doing my PhD at Aalborg University. My research work is finished, so I can start soon while I complete my thesis. What I like about Yuman is that you start from the nurses' needs. A Buddy that takes transport tasks off their hands gives them more time with patients, and it has to work safely around people every day. That is exactly the kind of robot I want to build.

In my PhD I designed a hybrid shoulder exoskeleton where a variable stiffness mechanism and a small 6 Nm motor work together to support the arm. I took it from hand sketches to SolidWorks, optimised the design with a genetic algorithm, integrated the motor, encoder and force sensors, measured its stiffness on a physical test setup and tested it with 12 people. It cut energy use by more than 25% and won the Best Paper Award at IFToMM ISRM 2026. I also designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms, estimated its stiffness with less than 10% error and quantified human arm impedance. When my first clutch design kept breaking and was hard to make, I replaced it with a simple bolt-locking clutch, which taught me to keep designs simple, buildable and reliable.

I would fit best in mechanical design and hardware: designing and improving parts of the Buddies, building and testing prototypes, integrating motors and sensors, and making the robots reliable for daily hospital use. I have not worked on mobile robot navigation or ROS yet, and I would be glad to learn that side from your team. I enjoy working close to users, as I did when testing my exoskeleton with participants, and I explain ideas quickly with hand sketches, since I also hold a Senior Diploma in Fine Arts.

I live in Denmark already and I am willing to learn Danish. Thank you for considering my application. You can see my work at arunabha22.github.io/arunabha-majumder.

Kind regards,
Arunabha Majumder